

Jiazi Bu

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Education

- PhD** **Nanyang Technological University**, College of Computing and Data Science Singapore
Sept 2025 - Present
- **Advisor:** Prof. [Xingang Pan](#).
 - **Programme:** *NTU-SJTU Joint PhD Degree Programme*. Fully funded.
- PhD** **Shanghai Jiao Tong University**, Computer Science and Technology Shanghai, China
Sept 2025 - Present
- **Advisor:** Prof. [Dahua Lin](#).
 - **Programme:** *NTU-SJTU Joint PhD Degree Programme*. Fully funded.
- BEng** **Shanghai Jiao Tong University**, Artificial Intelligence (*GuoZhi Class Programme*) Shanghai, China
Sept 2021 - June 2025
- **GPA:** 4.04/4.3; **Average Score:** 93.10/100; **Rank:** 5/95 (**TOP 5%**).
 - **Languages:** TOEFL iBT 105 (R:29, L:27, S:23, W:26); CET4: 636; CET6: 612.
 - **Related Coursework:** Mathematical Analysis (A+), Linear Algebra (A), Data Structure (A+), Data Mining (A+), Algorithm Design and Analysis (A+), Deep Learning (A+), Machine Learning (A), Reinforcement Learning (A+).

Experience

- MMLab @ NTU, S-Lab**, *PhD Student* Singapore
Sept 2025 - Present
- Research Fields: Generative AI.
 - Advisor: Prof. [Xingang Pan](#).
- Large Model Center @ Shanghai AI Laboratory**, *Undergraduate & PhD Research Intern* Shanghai, China
Nov 2023 - Present
- Research Fields: Generative AI and Large Vision-Language Models.
 - Mentors: [Jiaqi Wang](#), [Pan Zhang](#), [Xiaoyi Dong](#), [Yuhang Zang](#) and [Tong Wu](#).
- ReThinkLab @ SJTU**, *Undergraduate Research Intern* Shanghai, China
Oct 2022 - May 2024
- Research Fields: Computer Vision and AI4Science.
 - Advisor: Prof. [Junchi Yan](#).
- GuoZhi Class @ SJTU**, *Undergraduate Student (the first session)* Shanghai, China
Dec 2021 - June 2025
- Description: A training programme for top-notch talents in artificial intelligence, jointly established by Shanghai Jiao Tong University and Shanghai AI Laboratory.
 - Leaders: Prof. [Xiaoou Tang](#) and Prof. [Cewu Lu](#).

Publications (* for Co-First author)

[ICLR 2026] DiCache: Let Diffusion Model Determine Its Own Cache

Jiazi Bu*, Pengyang Ling*, Yujie Zhou*, Yibin Wang, Yuhang Zang, Tong Wu, Dahua Lin, Jiaqi Wang

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[NeurIPS 2025] HiFlow: Training-free High-Resolution Image Generation with Flow-Aligned Guidance

Jiazi Bu*, Pengyang Ling*, Yujie Zhou*, Pan Zhang, Tong Wu, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, Dahua Lin, Jiaqi Wang

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[CVPR 2025] ByTheWay: Boost Your Text-to-Video Generation Model to Higher Quality in a Training-free Way

Jiazi Bu*, Pengyang Ling*, Pan Zhang, Tong Wu, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, Dahua Lin, Jiaqi Wang

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[arXiv 2026] From Sparse to Dense: Multi-View GRPO for Flow Models via Augmented Condition Space

Jiazi Bu*, Pengyang Ling*, Yujie Zhou*, Yibin Wang, Yuhang Zang, Tianyi Wei, Xiaohang Zhan, Jiaqi Wang, Tong Wu, Xingang Pan, Dahua Lin

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[arXiv 2026] AdaGRPO: A Capability-Aware Adaptive Enhancement for Flow-based GRPO

Jiazi Bu*, Pengyang Ling*, Yujie Zhou*, Yibin Wang, Yuhang Zang, Tianyi Wei, Xiaohang Zhan, Jiaqi Wang, Tong Wu, Xingang Pan, Dahua Lin

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[ICLR 2025] MotionClone: Training-free Motion Cloning for Controllable Video Generation

Pengyang Ling*, **Jiazi Bu***, Pan Zhang, Xiaoyi Dong, Yuhang Zang, Tong Wu, Huaian Chen, Jiaqi Wang, Yi Jin

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[ICCV 2025] Light-A-Video: Training-free Video Relighting via Progressive Light Fusion

Yujie Zhou*, **Jiazi Bu***, Pengyang Ling*, Pan Zhang, Tong Wu, Qidong Huang, Jinsong Li, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, Anyi Rao, Jiaqi Wang, Li Niu

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[CVPR 2026] Fine-Grained GRPO for Precise Preference Alignment in Flow Models

Yujie Zhou*, Pengyang Ling*, **Jiazi Bu***, Yibin Wang, Yuhang Zang, Jiaqi Wang, Li Niu, Guangtao Zhai

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[ICME 2026] Uni-Classifier: Leveraging Video Diffusion Priors for Universal Guidance Classifier

Yujie Zhou*, Pengyang Ling*, **Jiazi Bu***, Bingjie Gao, Li Niu

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[arXiv 2026] Pave-GRPO: Beyond Instantaneous Guidance through Principled Average Velocity Decomposition

Pengyang Ling*, **Jiazi Bu***, Yujie Zhou*, Yibin Wang, Zhenyu Hu, Zihan Zhang, Yi Jin, Huaian Chen, Yuhang Zang

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[NeurIPS 2024] Unveiling The Matthew Effect Across Channels: Assessing Layer Width Sufficiency via Weight Norm Variance

Yiting Chen, **Jiazi Bu**, Junchi Yan

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[AAAI 2024] ViTree: Single-Path Neural Tree for Step-Wise Interpretable Fine-Grained Visual Categorization

Danning Lao, Qi Liu, **Jiazi Bu**, Junchi Yan, Wei Shen

[Paper](#) / [Code](#)

[SCIENCE CHINA Information Sciences] Drug-drug Interaction Prediction via Hierarchical Structure Modeling

Huaijin Wu, **Jiazi Bu**, Nianzu Yang, Yao Sun, Haitao Song, Ning Liu, Junchi Yan

[Paper](#) / [Code](#)

[arXiv 2026] Unified Personalized Reward Model for Vision Generation (a part of *UnifiedReward* series)

Yibin Wang, Yuhang Zang, Feng Han, **Jiazi Bu**, Yujie Zhou, Cheng Jin, Jiaqi Wang

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[arXiv 2025] UniGenBench++: A Unified Semantic Evaluation Benchmark for Text-to-Image Generation

Yibin Wang*, Zhimin Li*, Yuhang Zang*, **Jiazi Bu**, Yujie Zhou, Chunyu Wang, Qinglin Liu, Cheng Jin, Jiaqi Wang

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[arXiv 2025] Pref-GRPO: Pairwise Preference Reward-based GRPO for Stable Text-to-Image Reinforcement Learning (the unified T2I benchmark *UniGenBench* proposed in this work)

Yibin Wang*, Zhimin Li*, Yuhang Zang*, Yujie Zhou, **Jiazi Bu**, Chunyu Wang, Qinglin Liu, Cheng Jin, Jiaqi Wang

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[arXiv 2026] EndoCoT: Scaling Endogenous Chain-of-Thought Reasoning in Diffusion Models

Xuanlang Dai, Yujie Zhou, Long Xing, **Jiazi Bu**, Xilin Wei, Yuhong Liu, Beichen Zhang, Kai Chen, Yuhang Zang

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[arXiv 2026] CapRL++: Unified Reinforcement Learning with Verifiable Rewards for Dense Image and Video Captioning (the extended version of the ICLR 2026 paper *CapRL*)

Penghui Yang*, Long Xing*, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, Yibin Wang, Yujie Zhou, **Jiazi Bu**, Jianze Liang, Qidong Huang, Jiaqi Wang, Feng Wu, Dahua Lin

Projects

[\[arXiv 2023\]](#) **MetaScript: Few-Shot Handwritten Chinese Content Generation via Generative Adversarial Networks**

[Report 🔗](#) / [Code 🔗](#)

- **Jiazi Bu***, Qirui Li*, Kailing Wang*, Xiangyuan Xue*, Zhiyuan Zhang*
- Project of AI3604 Computer Vision, 2023 Fall, SJTU
- Developed a novel Chinese content generation system that can specify styles based on Generative Adversarial Networks (GANs).

Honors & Awards

Ye Jun and Shen Nanpeng Outstanding Scholarship	2025
• ¥ 20000, received as one of the TOP 30 graduates of SJTU.	
Zhiyuan Honors Scholarship	2021, 2022, 2023, 2024
• ¥ 5000 per year, received as part of the <i>SJTU Zhiyuan Honors Programme</i> .	
Guozhi Scholarship	2022
• ¥ 8000, received as being in the TOP 15% of SJTU AI students.	
SJTU Postgraduate Scholarship for PhD Student	2026
SJTU Undergraduate Scholarship	2022, 2023, 2024, 2025
CUMCM National First Prize (TOP 0.55%) & First Prize in Shanghai Chapter	2023
SJTU Zhiyuan Honor Bachelor's Degree	2025
SJTU President Award	2023
• Received for serving as a student volunteer during the COVID-19 pandemic.	
SJTU Outstanding Freshman Training Camp Member	2021
SJTU Outstanding Graduate	2025
SJTU Outstanding Bachelor's Thesis of School of Computer Science	2025
SJTU Best Bachelor's Thesis Award (TOP 1%)	2025

Services

Academic Reviewer

- Conference Reviewer: ECCV (2026), CVPR (2026), ICLR (2026), COLM (2026).
- Journal Reviewer: T-PAMI, TMLR.

Volunteer

- Student Volunteer: during the COVID-19 pandemic (for **55 hours**); CUMCM 2024.